Weight and Staging

Why most of a rocket is propellant, the one equation that links exhaust speed to final speed, and why dropping dead weight pays.

Isagi · November 2026 · Theory from NASA Glenn’s Beginner’s Guide to Rockets; ISRO figures checked 9 November 2026

HOW TO USE THIS CHAPTER

- **Print single-sided** and write in the margins.
- You need a calculator with a **ln** key (natural log). Do each example before reading the answer.
- Masses of the S200 (31 t dry) and L110 (9 t dry) are **public** figures; ISRO’s page gives propellant, the C25 dry mass and the lift-off mass. Results using them are estimates.

By the end of this chapter you can: state the ideal rocket equation and use it both ways; read a mass ratio as a propellant share; show with numbers why staging beats a single big stage; and estimate the speed change delivered by each LVM3 stage.

1 The problem: weight is the enemy

Chapter 0 showed that 86% of LVM3 is propellant and about 1.25% reaches orbit. NASA’s weight pages explain why. A rocket must carry everything it burns, so each kilogram of payload needs many kilograms of propellant, and each extra kilogram of tank needs more again. NASA groups the non-propellant mass into the **empty weight** (structure, engines, guidance, payload) and treats propellant as the rest. The question for this chapter is: given an exhaust speed, how much of the vehicle must be propellant to reach a target speed?

2 The ideal rocket equation

Newton's law for a body of changing mass is a momentum statement. In a short moment the rocket loses a small mass dm of propellant at exhaust speed v_e (relative to the rocket) and gains a small speed dv. Momentum balance gives $m \times dv = -v_e \times dm$. Add up the small steps (an integral, which you can take on trust) and you get the **ideal rocket equation**:

$$\Delta v = v_e \times \ln(m_0 / m_f)$$

Δv = speed gained
v_e = exhaust speed (Chapter 1: $I_{sp} \times g_0$)
m_0 = mass at start of the burn (everything, including propellant)
m_f = mass at end of the burn (propellant used up)
$R = m_0 / m_f$, the mass ratio

Three things to read from it. First, **Δv scales with v_e**, which is why Chapter 1 mattered. Second, **Δv grows only as the logarithm of the mass ratio**, so every extra bit of speed costs more propellant than the last. Third, the equation ignores gravity and drag; real flights need extra Δv to pay for them (Chapter 3).

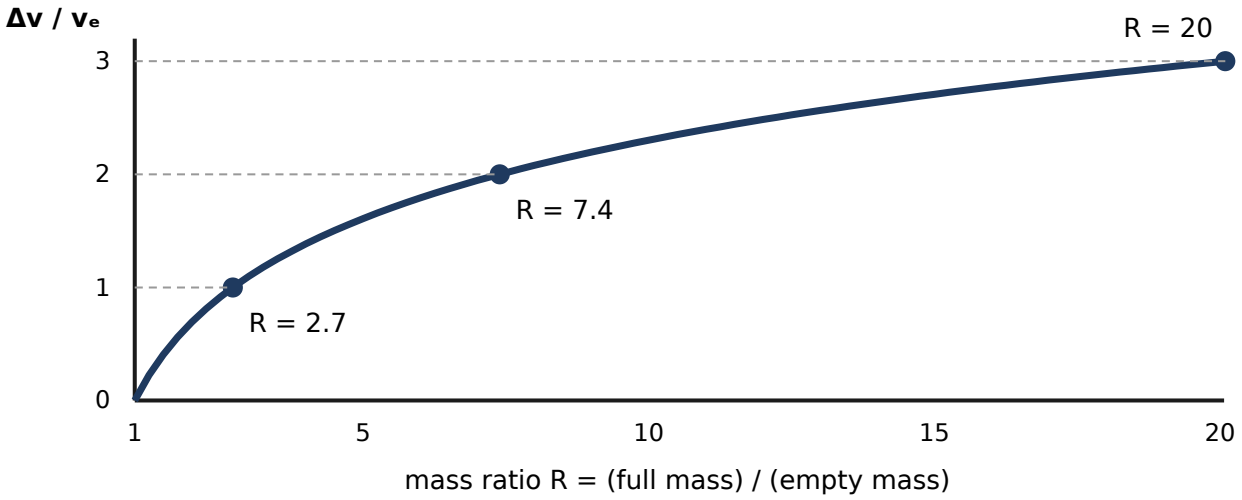


Figure 1. Δv in units of exhaust speed against mass ratio R. A mass ratio of 2.7 gives one v_e; to get three v_e you need R = 20. The curve flattens: this is the "tyranny" of the rocket equation.

Turn it around to get the mass ratio needed: $\mathbf{R = e^{\Delta v / v_e}}$. The share of the starting mass that must be propellant is $\mathbf{1 - 1/R}$. So $\Delta v = 1\ v_e$ needs 63% propellant, $2\ v_e$ needs 86%, $3\ v_e$ needs 95%.

ISRO LENS: A ROCKET CAN OUTRUN ITS EXHAUST

The C25's exhaust leaves at 4.3 km/s, yet LVM3 delivers a satellite at over 10 km/s. The equation shows how: speed is not limited by v_e; it needs a large mass ratio. The cost of speed shows up as propellant, not as a hard limit.

3 Staging: why it pays

A tank that is empty is dead weight, but a single stage keeps pushing it to the end. NASA explains the fix: drop the empty parts. Each later stage then pushes a lighter rocket, so m_f is smaller and $\ln(m_0/m_f)$ is larger. The speed gains of the stages simply **add**.

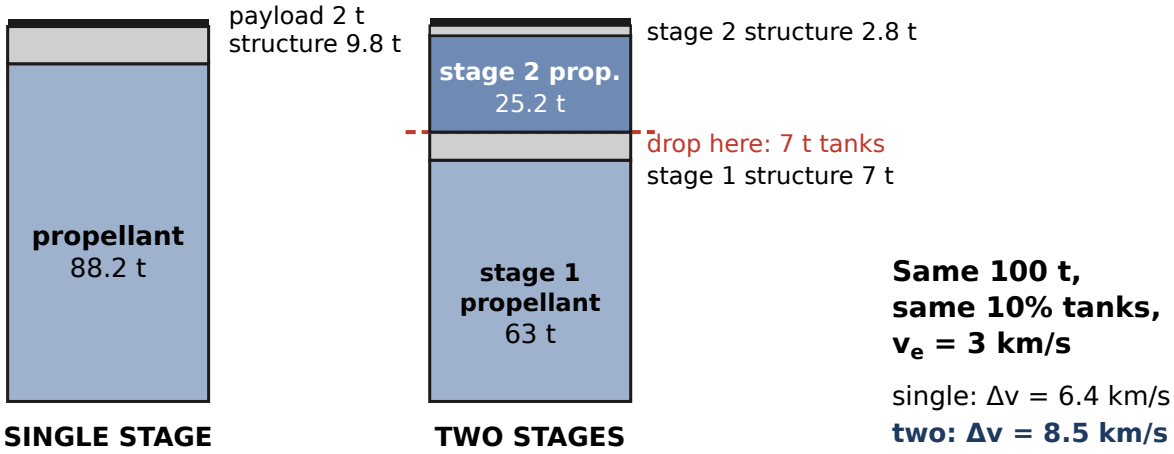


Figure 2. A toy comparison, drawn to scale in tonnes (not an ISRO vehicle). Dropping 7 t of empty tank halfway lifts Δv by a third. Worked in Example C.

The cost is complexity: more engines, more separation events and more things that can fail. That is why launch vehicles use two to four stages, not twenty.

4 Worked examples

All logarithms are natural (ln). $g_0 = 9.80665$ m/s². ISRO data: C25 28,000 kg propellant, 4,310 kg dry, GTO payload 4,000 kg, lift-off 640 t.

Example A: speed from the C25 alone

Assume the C25 burns with its 4 t GTO payload on top and ignore the fairing and equipment bay.

$$m_0 = 4{,}310 + 28{,}000 + 4{,}000 = 36{,}310 \text{ kg}$$
$$m_f = 4{,}310 + 4{,}000 = 8{,}310 \text{ kg}$$
$$R = 36{,}310 / 8{,}310 = 4.37$$
$$\Delta v = 4{,}344 \times \ln 4.37 = 4{,}344 \times 1.475 \approx 6{,}406 \text{ m/s}$$

One stage gives over 6.4 km/s because most of its mass is propellant: 28 of its 32.3 t, or 87%.

Example B: how much propellant for a given Δv?

Suppose you need $\Delta v = 4{,}000$ m/s using the C25's exhaust speed (4,344 m/s).

$$R = e^{4{,}000/4{,}344} = e^{0.921} \approx 2.51$$
$$\text{Propellant share} = 1 - 1/2.51 = 0.602 \rightarrow 60\%$$
$$\text{For the same job with the S200's } v_e \text{ (2,692 m/s):}$$
$$R = e^{4{,}000/2{,}692} = e^{1.486} \approx 4.42 \rightarrow 77.4\% \text{ propellant}$$

Higher exhaust speed cuts both the mass ratio and the propellant share. This is Chapter 1's I_{sp} turning into mass.

Example C: single stage versus two stages

Take a 100 t rocket with 2 t payload, tanks and engines 10% of each stage's mass, v_e = 3,000 m/s (Figure 2).

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Single stage: stage 98 t = 9.8 t structure + 88.2 t propellant
   R = 100 / 11.8 = 8.47   →   Δv = 3,000 × ln 8.47 ≈ 6,410 m/s

Two stages: stage 1 = 70 t (7 + 63), stage 2 = 28 t (2.8 + 25.2)
   Stage 1: R = 100 / 37  = 2.70  →  3,000 × 0.994 ≈ 2,980 m/s
   Stage 2: R =  30 / 4.8 = 6.25  →  3,000 × 1.833 ≈ 5,500 m/s
   Total ≈ 8,480 m/s   (+32%)
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Same propellant, same tank fraction, one third more speed. The only change is throwing away 7 t of empty tankage at the right moment.

Example D: Δv of each LVM3 stage (a rough estimate)

Treat LVM3 as three stages in series, with vacuum I_{sp} throughout. Use ISRO's 640 t, 205 t per booster, 115 t and 28 t, plus public dry masses: S200 31 t each, L110 9 t.

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Stage 1 (2 × S200):  m0 = 640 t,  mf = 640 − 410 = 230 t
   Δv = 2,692 × ln 2.78 ≈ 2,755 m/s
Drop boosters (62 t dry):  168 t.
Stage 2 (L110):      m0 = 168 t,  mf = 168 − 115 = 53 t
   Δv = 2,873 × ln 3.17 ≈ 3,315 m/s
Drop L110 (9 t dry):  44 t above it.
Stage 3 (C25):       m0 = 44 t,   mf = 44 − 28 = 16 t
   Δv = 4,344 × ln 2.75 ≈ 4,395 m/s
Total  ≈ 10,465 m/s
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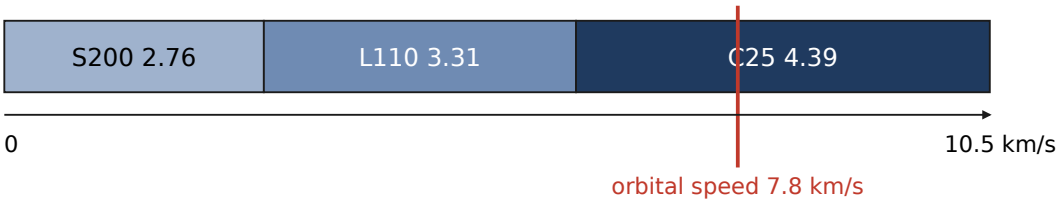


Figure 3. Estimated Δv per LVM3 stage (series model, vacuum I_{sp}). Scale: 50 px per km/s.

The total, about 10.5 km/s, is a little above the 9 to 10 km/s Chapter 0 gave for reaching orbit including losses. The gap is expected: vacuum I_{sp} overstates the ground-level push, and the model ignores the fairing and the core burning during the booster phase. Take it as a sanity check that the stage data hang together, not as an ISRO figure.

CHAPTER SUMMARY

- Ideal rocket equation: $\mathbf{\Delta v = v_e \times \ln(m_0/m_f)}$. Speed grows with exhaust speed and only with the logarithm of the mass ratio.
- Mass ratio needed: $R = e^{\Delta v/v_e}$; propellant share = $1 - 1/R$. $\Delta v = 3\ v_e$ already needs 95% propellant.
- A rocket can go faster than its own exhaust, at the price of mass ratio.
- Staging adds the speed gains of each stage and drops dead weight in between. In the toy case it gave +32% Δv.
- A rough LVM3 estimate gives about 4.4 km/s from the C25 and 10.5 km/s in total, in line with the 9 to 10 km/s needed.

Next, Chapter 3: motion and ascent. Gravity and drag losses, the shape of the climb, and how a vehicle turns upward flight into orbit.

Practice problems

Use $g_0 = 9.80665$ m/s² and a calculator with ln. Answers are on the last page.

P1. A stage has $v_e = 3{,}000$ m/s and a mass ratio of 5. What is its Δv?

P2. What mass ratio and propellant share does the C25 need for $\Delta v = 4{,}000$ m/s? (Use $v_e = 4{,}344$ m/s.)

P3. The C25 stage alone (propellant 28,000 kg, dry 4,310 kg), with no payload: find its mass ratio and Δv.

P4. In Example A, the payload grows from 4,000 kg to 5,000 kg. What is the new Δv, and how much has it dropped?

P5. Using the equation, explain in two sentences how a rocket can move faster than its own exhaust.

P6. Why do the Δv of separate stages add, even though each stage starts from a different mass?

Glossary

Empty weight (dry mass)
Everything except propellant: structure, engines, guidance, payload.

Propellant fraction
Propellant mass divided by starting mass.

Mass ratio (R)
Starting mass divided by final mass, m_0/m_f.

Δv (delta-v)
Change in speed a stage can deliver, in m/s.

Ideal rocket equation
$\Delta v = v_e \ln(m_0/m_f)$, valid without gravity or drag.

Natural logarithm (ln)
The log to base e (2.718…). ln(e) = 1.

Staging
Dropping spent stages so remaining engines push less mass.

Interstage
The structure joining two stages.

Payload fraction
Payload mass divided by lift-off mass.

Answer key

Fold this page back before you start the problems.

P1. $\Delta v = 3{,}000 \times \ln 5 = 3{,}000 \times 1.609 \approx 4{,}830$ m/s.

P2. $R = e^{4{,}000/4{,}344} = e^{0.921} \approx 2.51$. Propellant share $= 1 - 1/2.51 \approx 60\%$.

P3. $m_0 = 32{,}310$ kg, $m_f = 4{,}310$ kg, $R = 7.50$. $\Delta v = 4{,}344 \times \ln 7.50 = 4{,}344 \times 2.015 \approx 8{,}750$ m/s. That is the stage's limit with nothing on top; each tonne of payload eats into it.

P4. $m_0 = 37{,}310$, $m_f = 9{,}310$, $R = 4.007$; $\Delta v = 4{,}344 \times 1.388 \approx 6{,}030$ m/s. The drop is about 376 m/s for one extra tonne, a 5.9% loss.

P5. $\Delta v = v_e \times \ln R$ has no upper limit on Δv: if R is large enough, Δv exceeds v_e ($R = 20$ gives $3\ v_e$). Exhaust speed is measured relative to the rocket. Even when the rocket is faster than its exhaust speed, each kilogram thrown backwards still pushes it forward, so it keeps gaining speed while its mass falls.

P6. Each stage's equation measures speed gained during its own burn, from that stage's start mass to its end mass. Dropping a stage does not change the speed of what is left, so the next burn starts from the speed the last one reached, and the gains add.

Sources

Theory: NASA Glenn Research Center, Beginner's Guide to Rockets (weight, rocket equation, staging, empty weight). ISRO: LVM3 launch vehicle page (lift-off mass, propellant masses, C25 dry mass and payload). Public references: S200 and L110 dry masses and stage I_{sp} from the LVM3 Wikipedia entry, whose values differ slightly between sections; the series Δv model is the author's rough estimate, not an ISRO figure.